Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

Course Title: Basic Electrical and Electronic Circuit Lab

Course Code: EEE 0713 1102W

Experiment 06

Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit

---



Submitted By:
Raj Bhowmik Sany
Reg. No: 202561901048
Session: 2025
Department of Software Engineering

Submitted To:
Dr. Md Rasedujjaman
Associate Professor
Department of Electrical and Electronic Engineering
Shahjalal University of Science and Technology

Submission Date: August 31, 2026

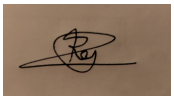
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Design and Performance Analysis of a Full-Wave Bridge Rectifier Circuit"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (EEE 0713 1102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: August 31, 2026



Raj Bhowmik Sany
Reg. No: 202561901048

Contents

Declaration **i**

List of Figures **iii**

List of Tables **iv**

Objective **1**

Introduction **1**

Theory **1**
Cycle-by-Cycle Conduction Analysis 1
Mathematical Formulation of Rectifier Parameters 1
Effect of a Capacitive Filter 2

Apparatus and Components **2**

Circuit Diagram **2**

Experimental Procedure **2**

Theoretical Calculation **3**

Data Collection **4**
Oscilloscope Waveforms 4

Result Analysis **5**

Conclusion **6**

References **6**

List of Figures

1 Full-wave bridge rectifier circuit: function generator feeding a four-diode bridge, with load R_L and an optional parallel filter capacitor C across the DC output. 3
2 Peak input waveform from the function generator ($V_m = 6\,\text{V}$, period 20 ms at 50 Hz). . 4
3 Unfiltered rectified output waveform: troughs sit at 0 V and peaks at the measured $V_{mo} = 2\,\text{V}$, ripple frequency 100 Hz. 5
4 Filtered output waveform: a mostly flat line near the calculated $V_{DC} = 1.78\,\text{V}$ with a small ripple riding on top, matching what was visibly seen on the DSO once the capacitor was connected. 5

List of Tables

1 List of Equipment and Components Used 2
2 Unfiltered Rectifier Performance 4
3 Filtered Rectifier Performance: Effect of the 47 µF Capacitor 4

Objective

- To design, configure, and construct a functional full-wave bridge rectifier circuit on a breadboard using four semiconductor diodes.
- To analyze the cycle-by-cycle conduction mechanics of the bridge topology during both the positive and negative half-cycles of the input AC signal.
- To capture and compare the input and output voltage waveforms of the rectifier using a Digital Storage Oscilloscope (DSO).
- To measure and compute key performance parameters, including the average DC output voltage (V_{DC}), RMS output voltage (V_{rms}), ripple factor (r), and rectification efficiency (η).
- To analyze the effect of a parallel capacitive filter on output smoothing, and to compare all measured results against theoretical predictions.

Introduction

Most electronic circuits need a steady DC supply to work properly, but the power that comes out of a wall socket is AC. A rectifier is the circuit that bridges that gap: it takes an alternating input and turns it into a one-directional, pulsating output. There are a few ways to build one, but the full-wave bridge rectifier is probably the most common in practice, mainly because it makes use of both halves of the AC cycle and doesn't need a center-tapped transformer the way some older designs do. It manages this with just four diodes arranged in a diamond pattern – no matter which half of the input cycle it's in, the current through the load resistor always ends up flowing the same way. On its own the output is still pretty rough, more of a bumpy DC than a flat line, so in this experiment we also look at what happens when a capacitor is added in parallel with the load to smooth things out.

Theory

Cycle-by-Cycle Conduction Analysis

Suppose the AC source feeding the bridge varies with time as

$$v_s(t) = V_m \sin(\omega t),$$

where $V_m = V_{rms}\sqrt{2}$ is the peak secondary voltage and $\omega = 2\pi f$ is the angular frequency of the AC mains.

- **Positive Half-Cycle ($0 < \omega t < \pi$):** The top secondary terminal is positive with respect to the bottom one. This forward-biases D_1 and D_4, while D_2 and D_3 stay reverse-biased and effectively act as open switches. Current leaves the source, passes through D_1, goes down through the load R_L, and returns to the source through D_4.
- **Negative Half-Cycle ($\pi < \omega t < 2\pi$):** Once the input flips polarity, the roles swap: D_2 and D_3 turn on while D_1 and D_4 turn off. The current still flows through R_L in the same top-to-bottom direction as before – it's just routed through D_2 and D_3 instead.

Since the load current points the same way in both halves of the cycle, the output ends up with two ripples for every one cycle of the input, i.e. $f_{\text{out}} = 2f_{\text{in}}$.

Mathematical Formulation of Rectifier Parameters

Two diodes are always conducting at once, so their combined forward drop ($V_D \approx 0.7\,\text{V}$ each) has to be subtracted from the peak input before we get the actual peak output voltage:

$$V_{mo} = V_m - 2V_D.$$

From there, the usual full-wave rectifier results give the DC (average) output voltage and the RMS output voltage:

$$V_{DC} = \frac{2V_{mo}}{\pi}, \qquad V_{rms} = \frac{V_{mo}}{\sqrt{2}}.$$

The ripple factor tells us how much AC "leftover" is still riding on top of the DC value. For an unfiltered full-wave rectifier it comes out to

$$r = \sqrt{\left(\frac{V_{rms}}{V_{DC}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} \approx 0.482$$

and notably this doesn't depend on the specific voltage or load chosen – it's a fixed property of the topology itself. Two more numbers matter for evaluating the rectifier: the rectification efficiency, and the Peak Inverse Voltage (PIV) each diode needs to withstand:

$$\eta = \frac{P_{DC}}{P_{AC}} \times 100\%, \qquad \eta_{\max} = \frac{8}{\pi^2} \approx 81.2\%, \qquad \text{PIV} = V_m.$$

Effect of a Capacitive Filter

Adding a capacitor C across the load helps smooth things out. It charges up close to V_{mo} at each peak, then discharges slowly through R_L until the next peak comes along, so the output no longer drops all the way back to zero. If we assume the discharge is roughly linear (a fair approximation once the ripple is small), the peak-to-peak ripple voltage and ripple factor for a full-wave rectifier (ripple frequency $2f$) work out to

$$V_{r(p\text{-}p)} \approx \frac{I_{DC}}{2fC} = \frac{V_{DC}}{2fR_LC}, \qquad r \approx \frac{1}{4\sqrt{3}\, fR_LC}.$$

Both formulas point the same way: increasing either C or R_L brings the ripple down. The trade-off is that a larger capacitor also pulls a bigger surge of inrush current through the diodes each time it recharges near the peak.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	**Specification / Rating**	**Quantity**
Function Generator	Sine wave, 50 Hz, measured peak output 6 V	1
Silicon Rectifier Diodes	1N4007	4
Load Resistor R_L	965 Ω	1
Electrolytic Filter Capacitor C	47 µF	1
Dual-Channel Digital Storage Oscilloscope (DSO)	With 10× probes	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

Note: The function generator's output level was read directly off the DSO rather than off a front-panel dial.

Circuit Diagram

Experimental Procedure

1. **Bridge Assembly:** Place the four 1N4007 diodes on the breadboard in the diamond/bridge layout shown in Figure 1. Double-check the anode/cathode orientation on each one before moving on – getting a diode backwards here is an easy mistake to make.

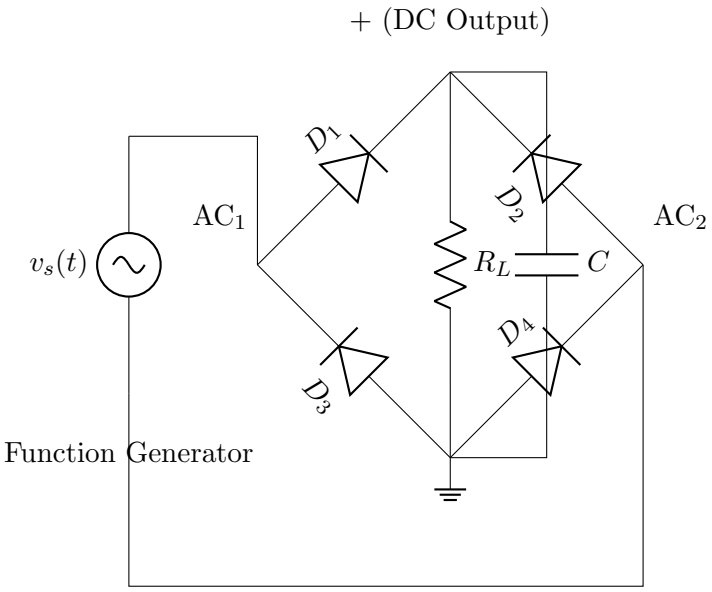


Figure 1: Full-wave bridge rectifier circuit: function generator feeding a four-diode bridge, with load R_L and an optional parallel filter capacitor C across the DC output.

2. **Source Integration:** Wire the output leads of the function generator into the two AC input nodes of the bridge (AC_1 and AC_2), set to a 50 Hz sine wave.

3. **Unfiltered Baseline Testing:** With no filter capacitor connected yet, put $R_L = 965\,\Omega$ across the DC output terminals. Show the AC input on DSO Channel 1 and the (still pulsating) DC output on Channel 2, and read the peak input V_m and peak output V_{mo} directly off the DSO trace. V_{DC}, V_{rms}, the ripple factor, and the efficiency are then calculated from V_{mo} using the formulas in the Theory section, since no separate way to measure them directly was demonstrated for this experiment, and no DMM was used.

4. **Filtered Analysis:** Add the $C = 47\,\mu F$ filter capacitor in parallel with the same R_L – make sure the polarity is right, since it's electrolytic. The output visibly smooths out on the DSO; V_{DC} and the peak-to-peak ripple $V_{r(p\text{-}p)}$ for this stage are likewise calculated from the measured V_{mo} using the filter formulas from the Theory section.

Theoretical Calculation

This section uses the real numbers from the bench: a measured peak input $V_m = 6\,V$, a diode drop of $V_D = 0.7\,V$, mains frequency $f = 50\,Hz$, and the load actually used, $R_L = 965\,\Omega$.

$$V_{mo} = V_m - 2V_D = 6 - 1.4 = 4.60\,V,$$

$$V_{DC} = \frac{2V_{mo}}{\pi} = \frac{2 \times 4.60}{3.1416} = 2.928\,V, \qquad I_{DC} = \frac{V_{DC}}{R_L} = \frac{2.928}{965} = 3.03\,mA,$$

$$V_{rms(\text{out})} = \frac{V_{mo}}{\sqrt{2}} = \frac{4.60}{1.4142} = 3.253\,V,$$

$$\eta = \frac{V_{DC}^2/R_L}{V_{rms(\text{out})}^2/R_L} \times 100\% = \frac{2.928^2}{3.253^2} \times 100\% \approx 81.0\%, \qquad \text{PIV} = V_m = 6\,V.$$

This 81.0% efficiency lines up with the fixed theoretical maximum $\eta_{\max} = 8/\pi^2 \approx 81.2\%$ from the Theory section, which is a good internal check that this part of the calculation is being done correctly. With the 47 µF filter capacitor added,

$$V_{r(p\text{-}p)} \approx \frac{V_{DC}}{2fR_LC} = \frac{2.928}{2 \times 50 \times 965 \times 47 \times 10^{-6}} \approx 0.646\,V.$$

These are the numbers the circuit *should* produce if it behaves like the idealized model in the Theory section. How they actually compare to what was measured is discussed next.

Data Collection

Table 2: Unfiltered Rectifier Performance

Load R_L	**Peak Input** V_m	**Peak Output** V_{mo}	**Measured** V_{DC}	**Measured** V_{rms}	**Ripple Factor** r	**Efficiency** η
965 Ω	6 V	2 V	1.27 V	1.41 V	0.483	81.1%

Note: V_m and V_{mo} are read directly off the DSO. V_{DC}, V_{rms}, the ripple factor, and the efficiency are calculated from V_{mo} using the formulas in the Theory section, since no separate way to measure them was demonstrated.

Table 3: Filtered Rectifier Performance: Effect of the 47 µF Capacitor

Load R_L	**Filter** C	**Measured** V_{DC}	**Ripple** $V_{r(p\text{-}p)}$
965 Ω	47 µF	1.78 V	0.44 V

Note: V_{DC} and $V_{r(p\text{-}p)}$ for the filtered stage are calculated from the measured V_{mo} using the filter formulas in the Theory section. The output visibly flattened out on the DSO once the capacitor was connected, consistent with this.

Oscilloscope Waveforms

The input, unfiltered output, and filtered output waveforms captured during the experiment are shown below.

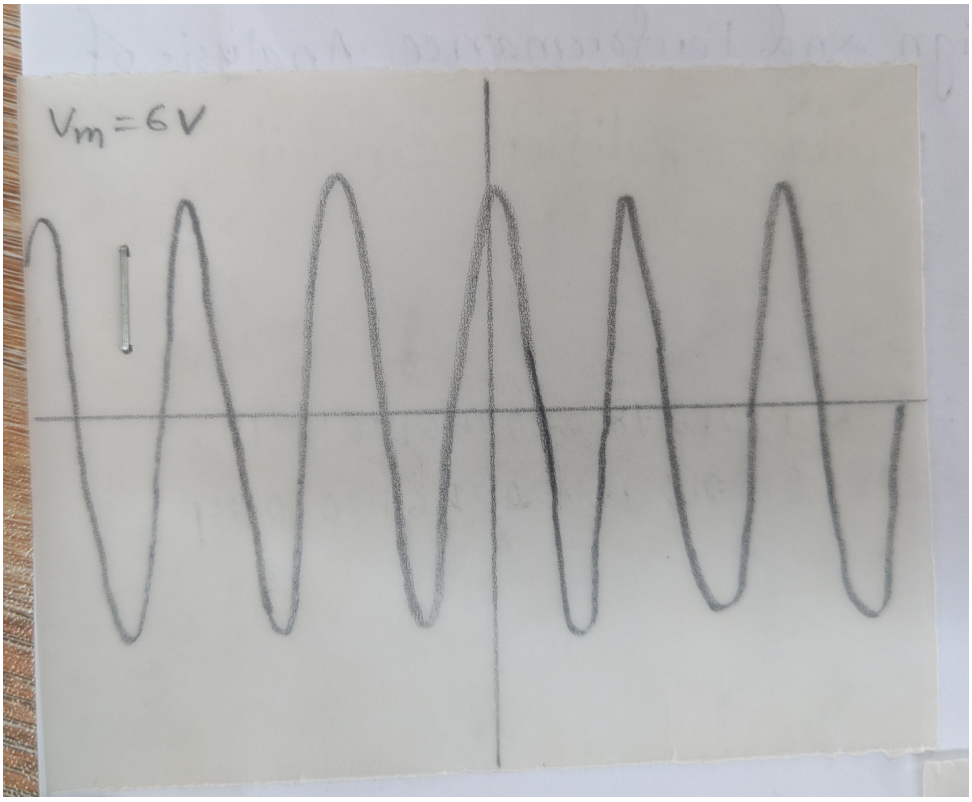


Figure 2: Peak input waveform from the function generator ($V_m = 6$ V, period 20 ms at 50 Hz).

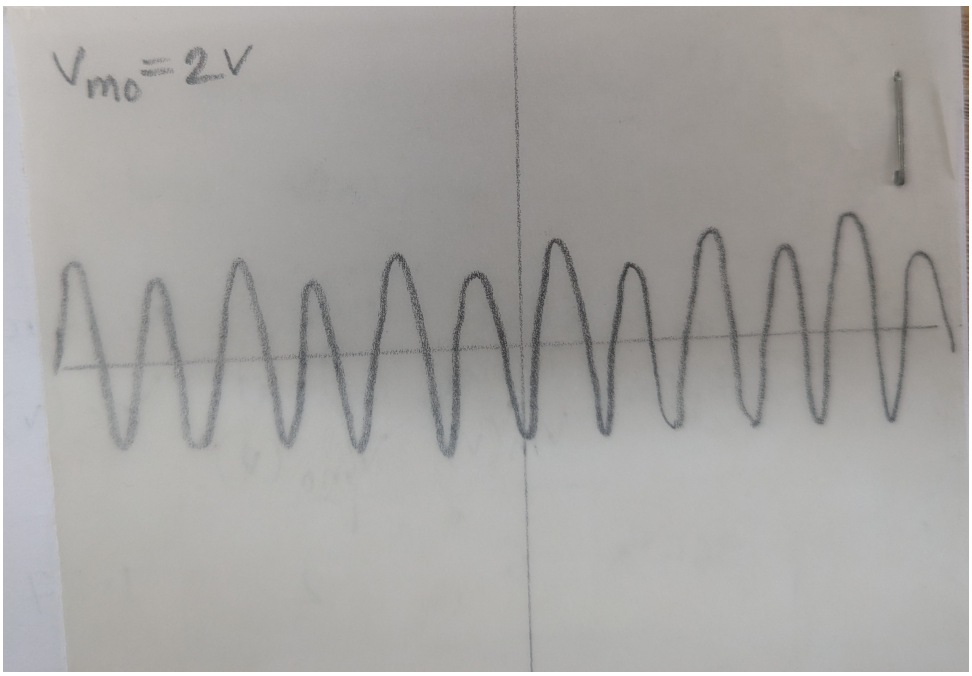


Figure 3: Unfiltered rectified output waveform: troughs sit at 0 V and peaks at the measured $V_{mo} = 2\,\text{V}$, ripple frequency 100 Hz.

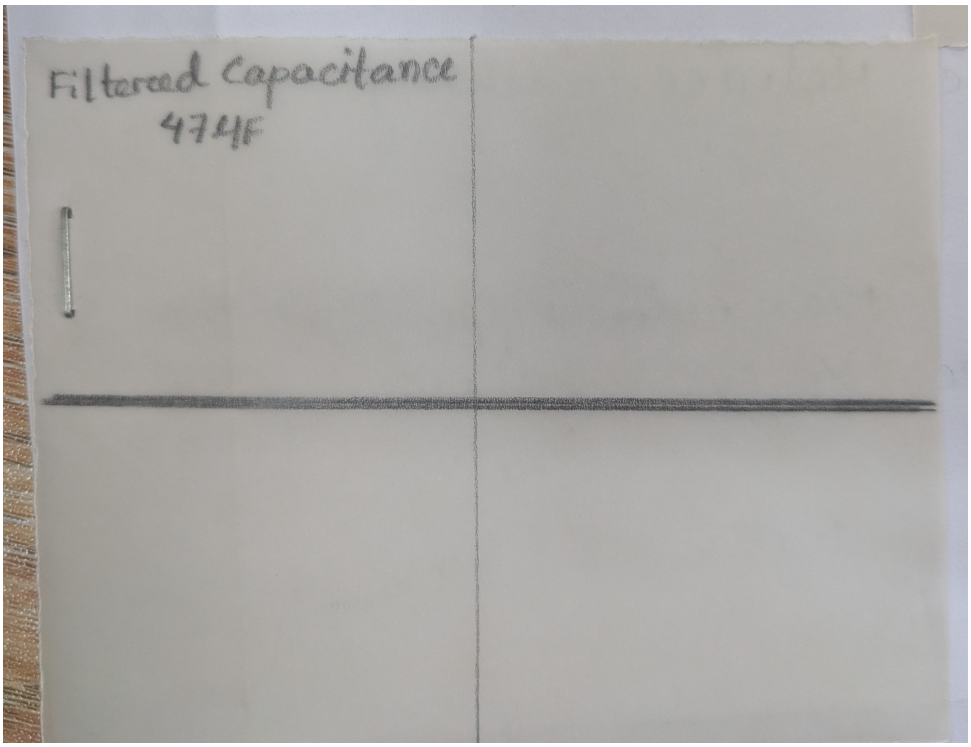


Figure 4: Filtered output waveform: a mostly flat line near the calculated $V_{DC} = 1.78\,\text{V}$ with a small ripple riding on top, matching what was visibly seen on the DSO once the capacitor was connected.

Result Analysis

The measured peak output, $V_{mo} = 2\,\text{V}$, comes in noticeably below the 4.60 V predicted in the Theoretical Calculation, about a 56% gap. Since V_{DC}, V_{rms}, and I_{DC} in Table 2 are all calculated directly from V_{mo}, that same gap carries through to all of them. What lines up much better with theory are the two quantities that only depend on the shape of the waveform rather than its size: the ripple factor came out to 0.483 against a fixed theoretical value of 0.482, and the efficiency came out to 81.1% against the

fixed theoretical maximum of 81.2%. Both are within about a quarter of a percent of theory, which is a good sign that the circuit is genuinely rectifying both halves of the input cycle correctly – it is really just the absolute voltage level that came out lower than the simple $V_m - 2V_D$ estimate. After the 47 µF filter capacitor was added, the calculated V_{DC} rises from 1.27 V to about 1.78 V and the ripple drops from 2 V peak-to-peak down to about 0.44 V, which is the expected direction and matches the smoother trace seen on the DSO once the capacitor was connected.

Conclusion

The full-wave bridge rectifier was assembled and tested with a function generator supplying $V_m = 6$ V at 50 Hz, $R_L = 965\,\Omega$, and a 47 µF filter capacitor. The peak input and peak output were read directly off the DSO; V_{DC}, V_{rms}, the ripple factor, and the efficiency were then calculated from the measured peak output using the standard full-wave rectifier formulas, since no separate way to measure them was shown for this experiment. The ripple factor and efficiency came out very close to their fixed theoretical values, confirming the circuit rectified correctly, even though the peak output itself was lower than the simple diode-drop estimate predicted. Adding the filter capacitor raised the calculated V_{DC} from 1.27 V to about 1.78 V and reduced the ripple, consistent with what a working filter should do.

References

1. Boylestad, R. L., & Nashelsky, L. (2013). *Electronic Devices and Circuit Theory* (11th ed.). Pearson.
2. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education.
3. Sedra, A. S., & Smith, K. C. (2014). *Microelectronic Circuits* (7th ed.). Oxford University Press. (Chapter 3: Diodes).